

Session 09 // objective

①

- ☐ identify Scaling bottlenecks in GenAI application, including LLM latency, token usage and API rate limits.
 - ☐ Apply Scaling and optimisation technique
 - ☐ Introduction to K8n. → S-6H
 - ☐ Deploy and operates a containerised AI application, demonstrating Scaling, monitoring, updates and roll back.
 - ☐ Design on e2E GenAI deployment architecture, connecting APIs
- (end 2 eval)
- Wrong | Right X

②

(10 Request)

(5 sec)
Response

Request
(0 sec)

(waiting)

⑧



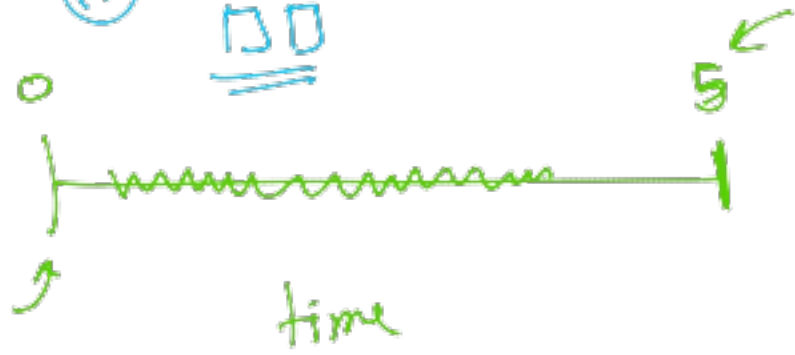
⑩

(server)

Fast API

⑩

100



⑩ (slot)

2

2

/chat



✓ Latency → time taken to process one input

✓ Concurrency

: at a moment → How many request server

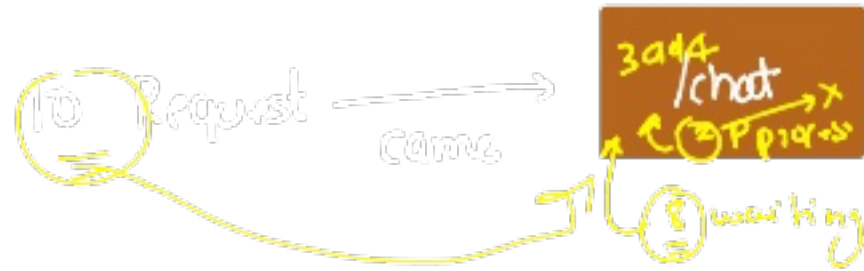
✓ Throughput

: in a given interval time → How many Request

$$\frac{2}{5} \Rightarrow \underline{\underline{0.4}}$$

↗

③



① Assume Avg 5 sec process any point

② Assume 2 request maximum request at a time

Request	Process time	Response arrive	Total latency since an
✓ ✓ 1 & 2	0	5	5
3 and 4	5	10	10
✓ 5 and 6	10	15	15

** Total latency = waiting time in queue + processing time

✓ Help me?

✓ Q177

Req 9 and 10 → what will be response time!

2

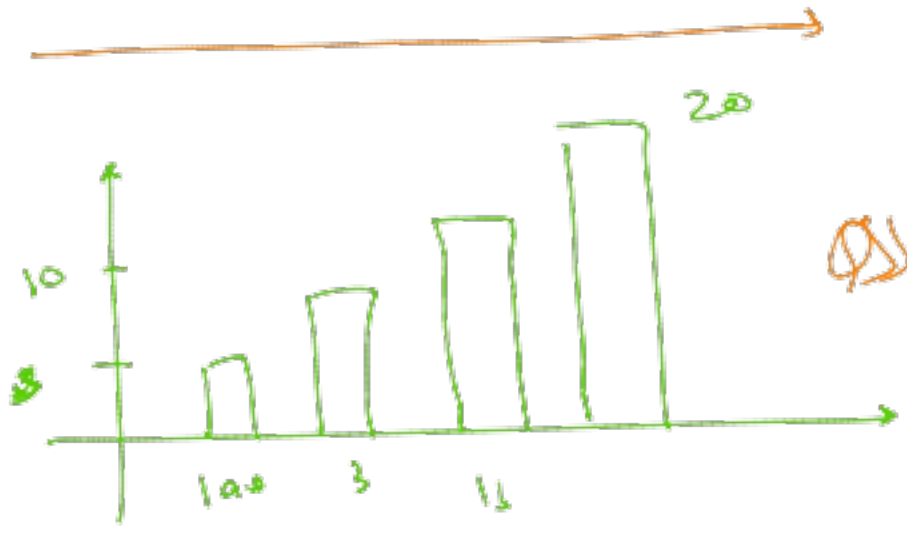


✓ Q277

in previous of what is maximum processing time?

✓ Q377

if in 5 sec we can complete 2 req then what is avg throughput?



$$\frac{2}{5 \text{ sec}} \Rightarrow 0.4 \text{ req/sec}$$

(1-2) - 5s
3 - 10
5-6 - 15
7-8 - 20
9-10 - 25 ✓

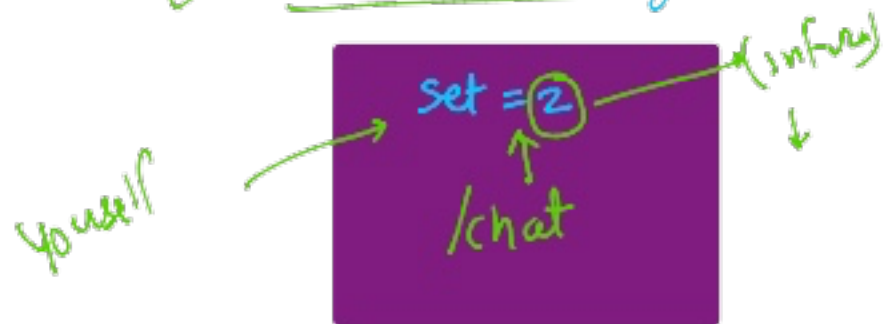
Q3 in 5 sec
10 Req
→ Thr = ?
2 ✓

General Question?

⑤

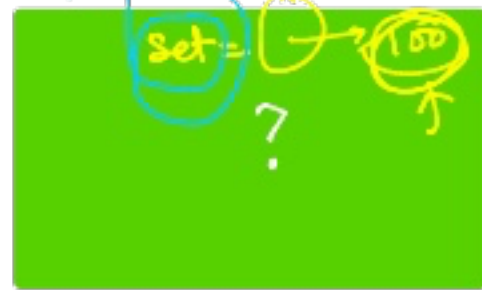
How you know that your system can handle 234 req/s at the same time

✓ 1) Explicit setting



Summary

(9 sec)



At End → Task → ① avg latenc time

(Break)

** Lets See Working Demo **

(ollama)

↓
(dora)

↔

(dora)

(LLM inference)

Some

circumstances

↳ if we increase the limit from 2 to 20 → will application
10 times faster?



eg →

↳ next page

max = 2 → 10
LM provide
OK
Any sen

Why?

(Analysis)

Reason?

time



Component

✓ ①

✓ Application

✓ ②

LUM Service

closed →

Open etc

③

RAN

②

Question

10

10

Capacity

maximum 10 Request

maximum 2 call

slot = ② ⑤ ⑩ 100

4u



com

(2) Request | at | fi

(100)

↑

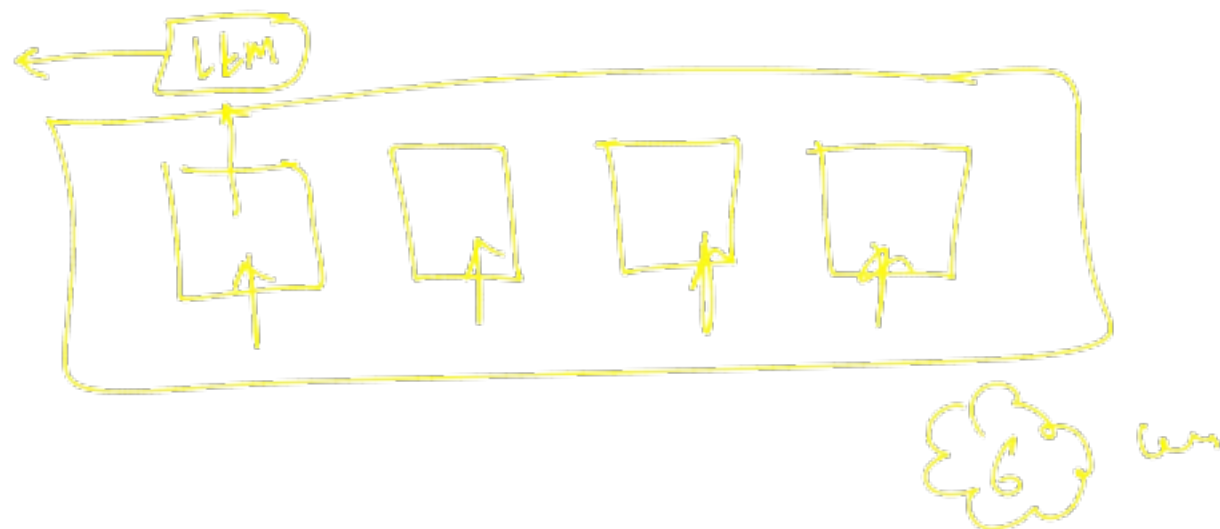
RAN

server

R10

2)

CDU

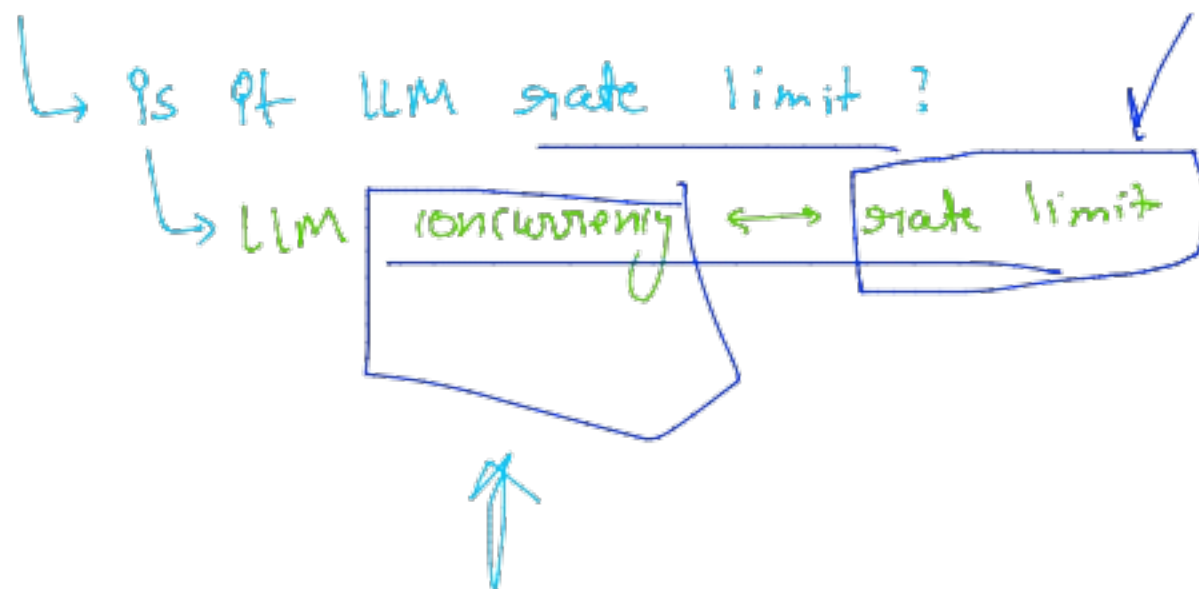


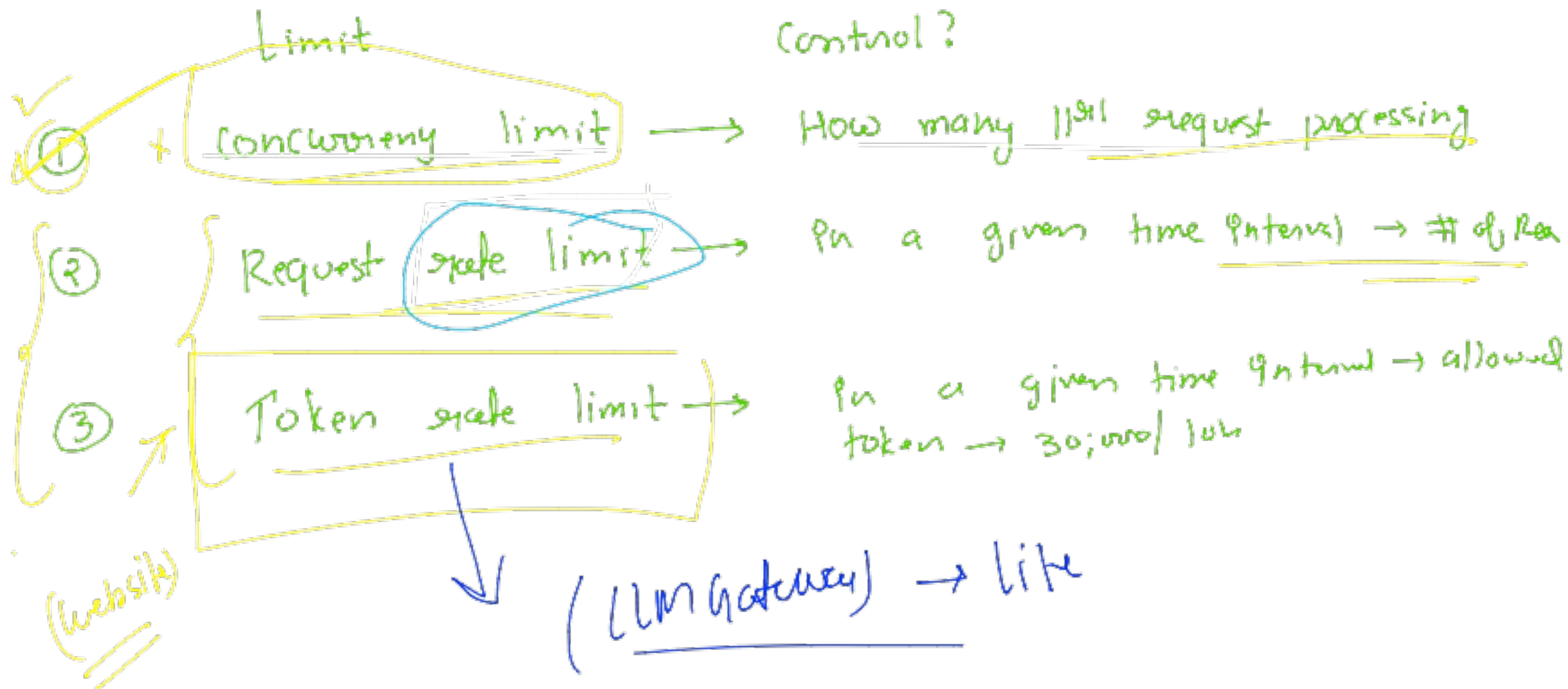
10 Request

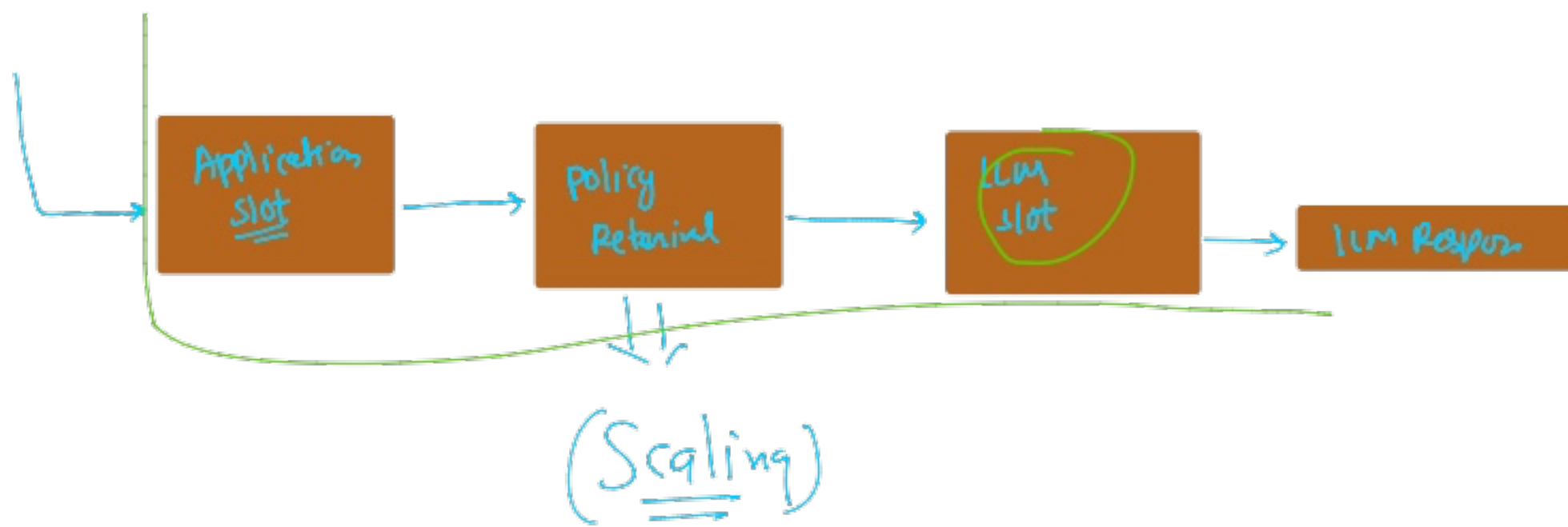
[lets see working demo]

learning → Identify where is the problem → Increase the capacity

often confusion







Scaling → Increasing Application capacity so that we can handle more request

① Vertical Approach → RAM / mem / CPU



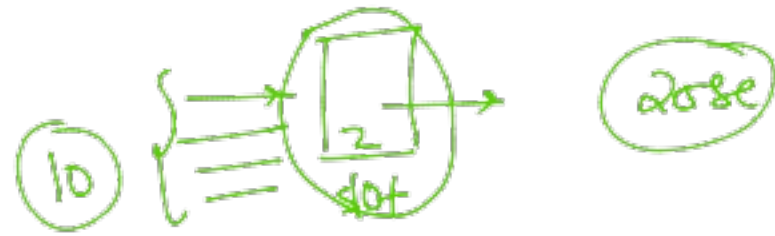
↳ ?

(code)
↓

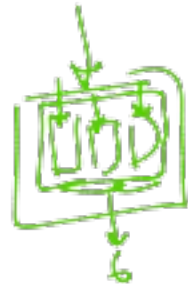
↑
(RAM)
↑
(data)
↑
proc ✓

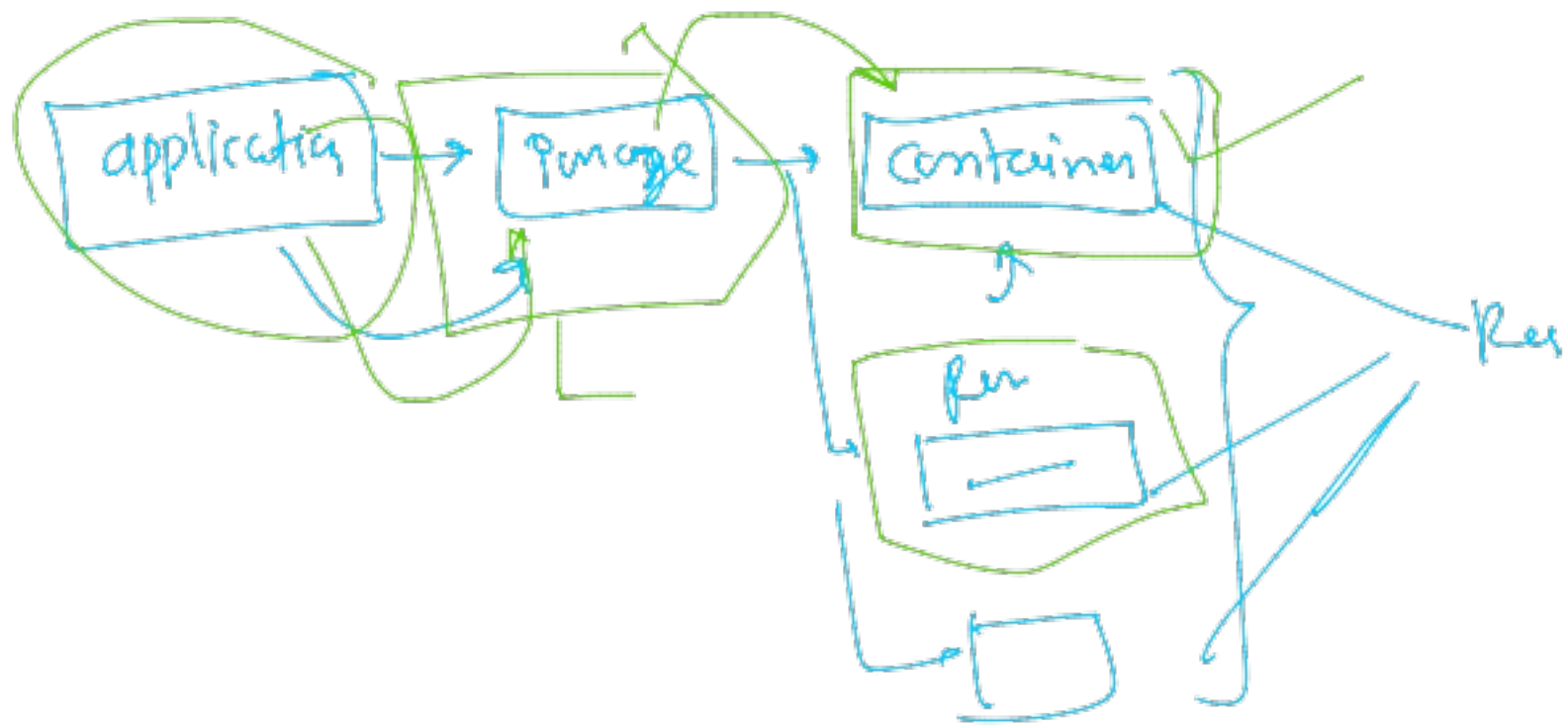
↳ ② Horizontal Scating

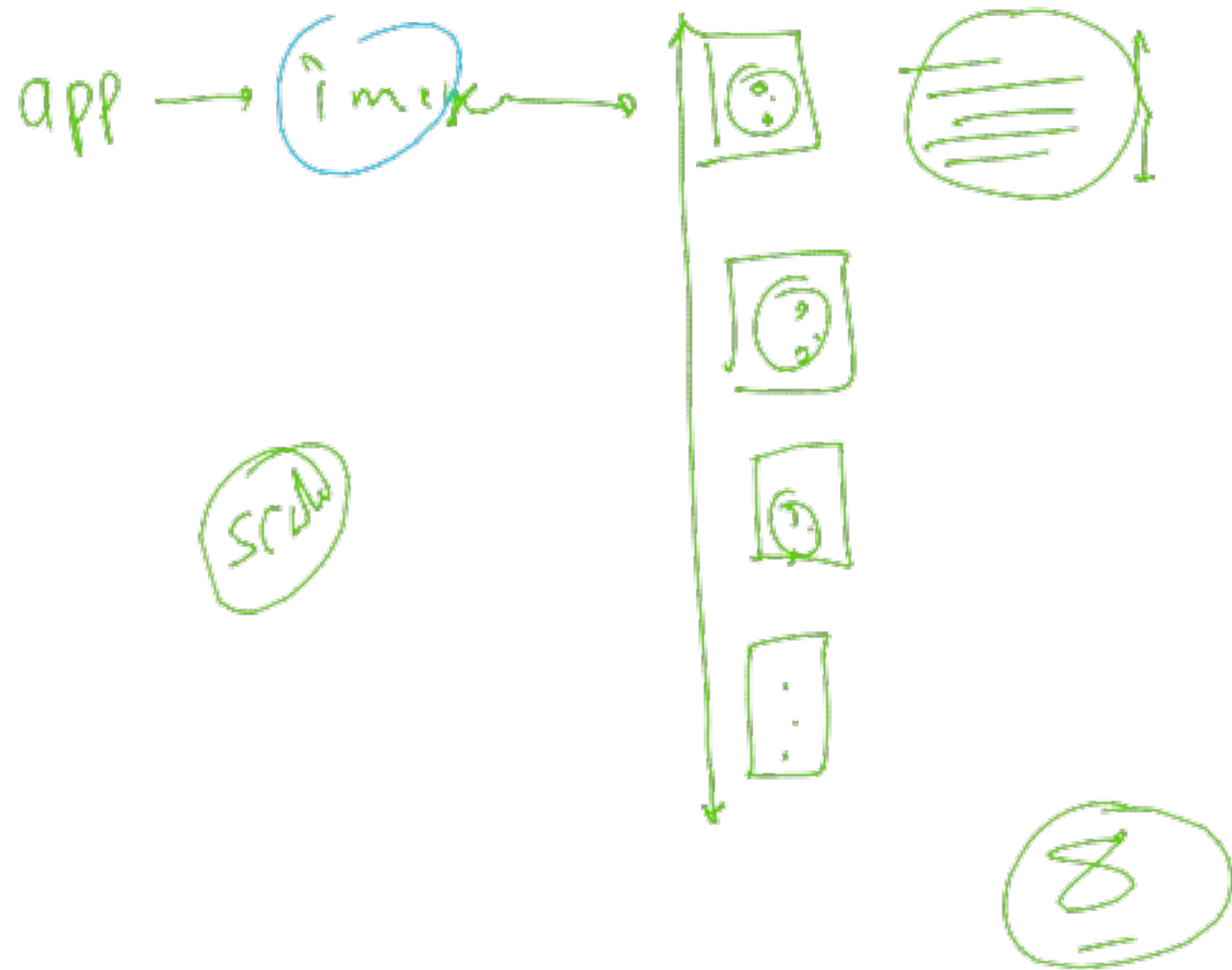
avg $\rightarrow$ (5 sec)

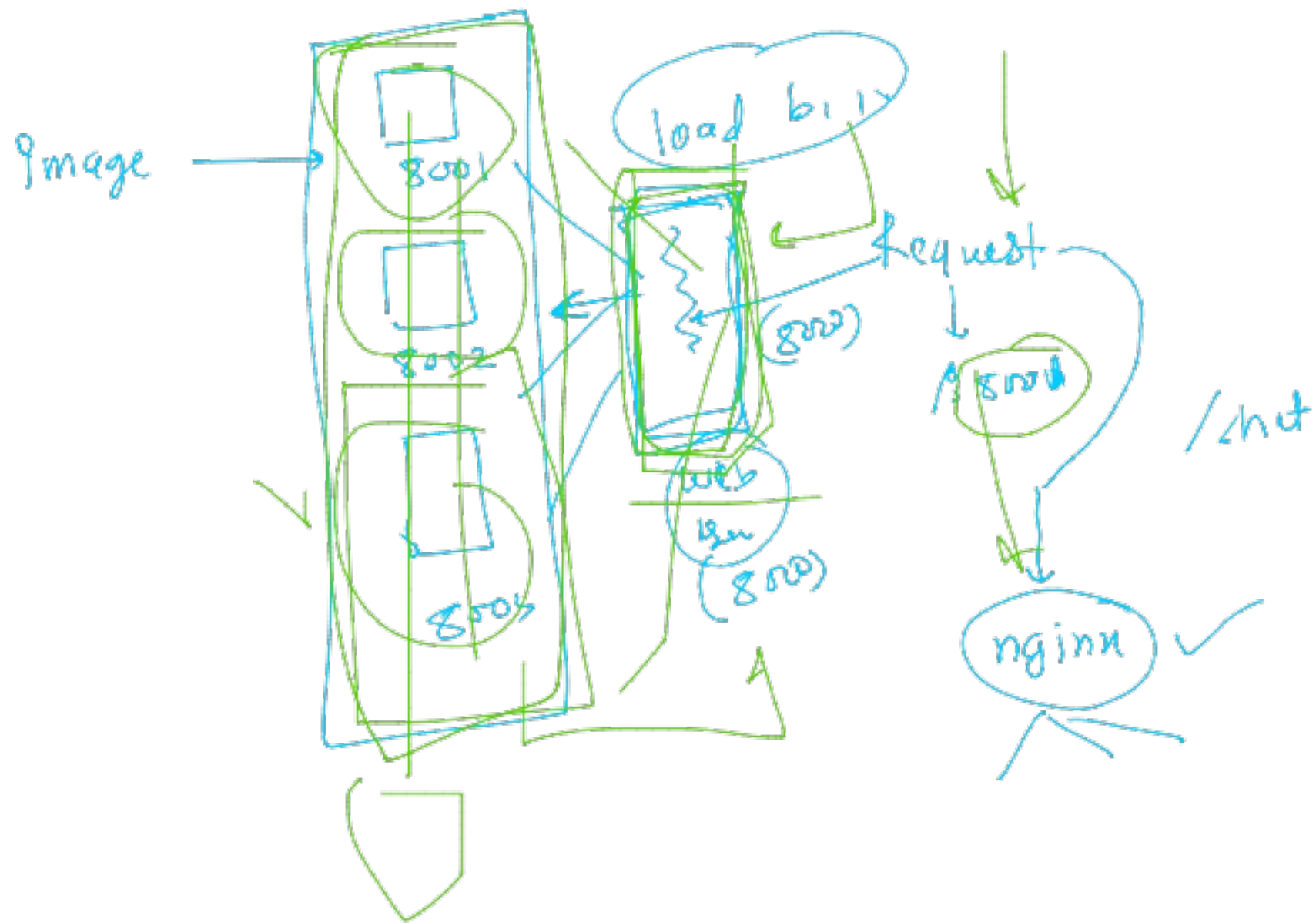


Setup	Total available processing slot	ideal throughput
✓ ① <u>instance</u> $\rightarrow$	②	? $2 \div 5 \Rightarrow 0.4$
2 <u>instance</u> $\rightarrow$	4	? $4 \div 5 \Rightarrow$ (0.8)
3 <u>instance</u> $\rightarrow$	6 ✓	? $6 \div 5 \Rightarrow$ (1.2)

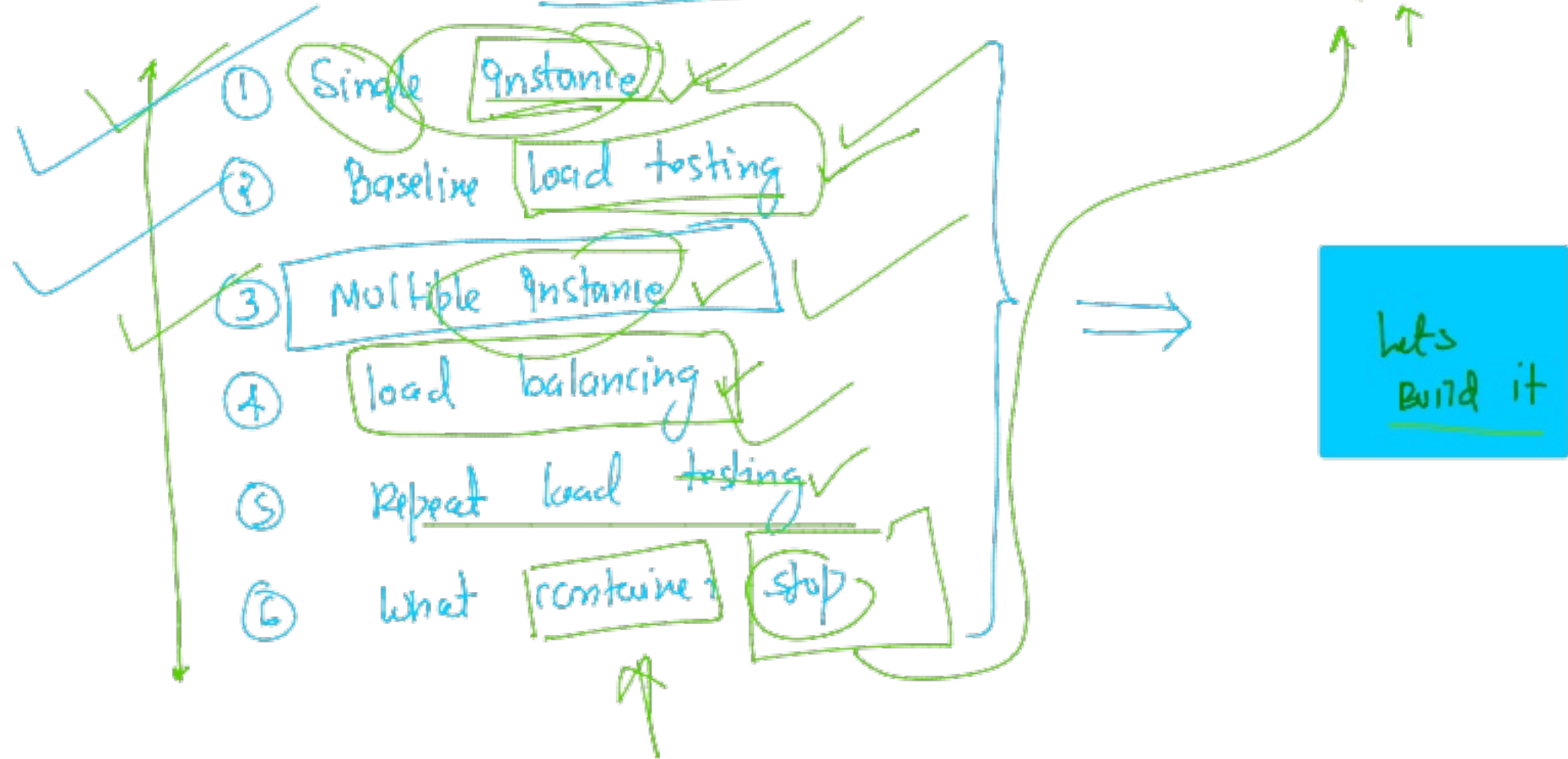


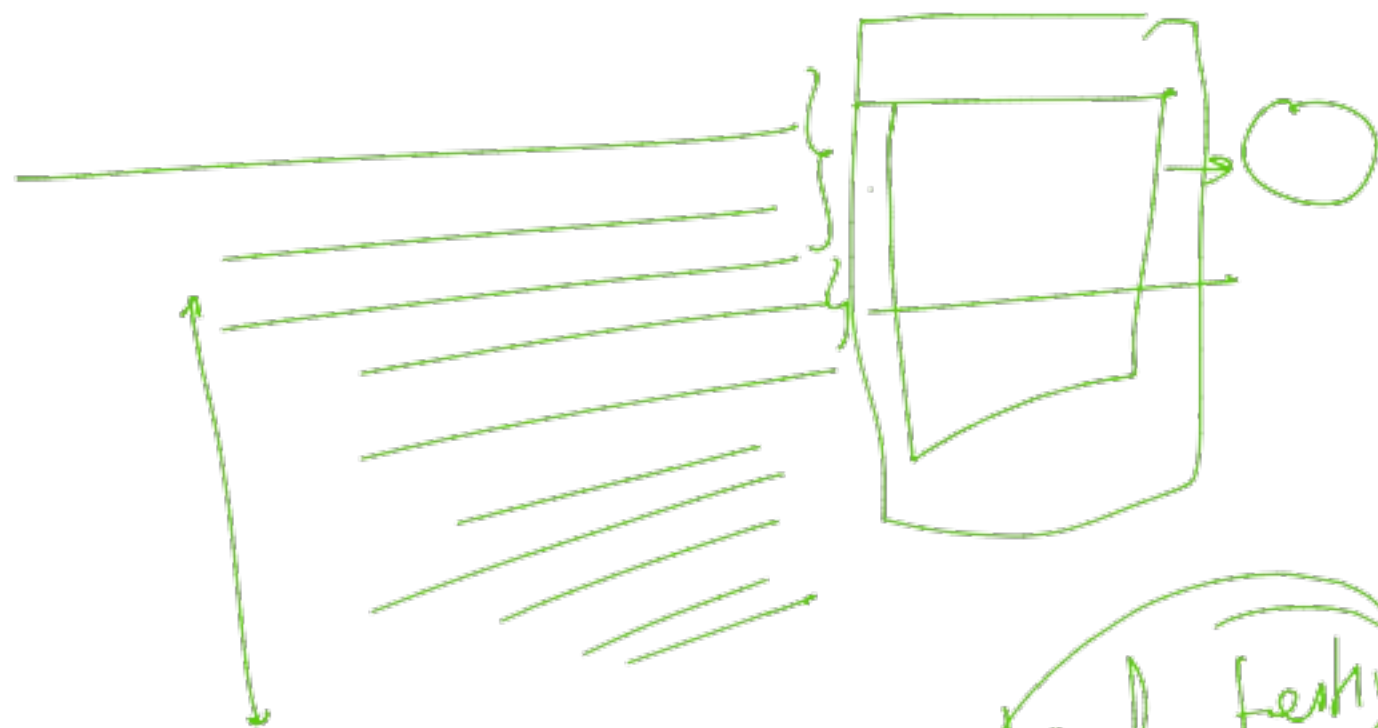






Hands-on // Plan of Action → (k n 8)





load test

→ 100